Problem Set 10

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: UDL 3–4 (shallow and deep networks, activations); UDL 6–7 (fitting models, gradients, backpropagation). Problems marked **[code]** should be submitted as a program in a language of your choice (Julia or Python recommended). The cited problem numbers are from UDL (Prince, 2023).

1. UDL, Chapter 3 problems: any two on shallow networks (e.g. Problems 3.1–3.2).
2. UDL, Chapter 4 problems: any two on deep networks and activation functions (e.g. Problems 4.1–4.2).
3. UDL, Chapter 7 problems: any two on gradients and backpropagation (e.g. Problems 7.1–7.2).
4. **[code]** Implement backpropagation for a network with one hidden layer on a small regression task. Verify your gradients against finite differences, then train with gradient descent and plot the loss curve.
5. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).